

The July 2026 Model Wave — Under the Hood

Day 99 marks a course-correction: after 22 industry verticals, this series returns to what it was built for — teaching new AI technology in depth. The timing could not be better, because the biggest model wave since GPT-5 is breaking right now. In five weeks: **Claude Sonnet 5** shipped with a native 1M-token context and a new tokenizer, **Claude Fable 5** came back from a 19-day safety suspension behind a classifier-gated architecture, **OpenAI** unveiled the **GPT-5.6 family (Sol, Terra, Luna)** in a government-vetted preview, and **Gemini 3.5 Pro** slipped to a July window with a 2M-token context. Today: what actually changed under the hood, the long-horizon agentic RL technique powering it all, and how to rebuild the 70/25/5 routing stack around the new tiers.

1M tokens — Sonnet 5's native context; default and only size	19 days Fable 5 offline (Jun 12–Jul 1) before safeguarded redeploy	~20 orgs government-vetted partners in the GPT-5.6 preview	750 tok/s GPT-5.6 Sol served on Cerebras hardware this July
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1. Claude Sonnet 5 — the 1M-token workhorse (and a tokenizer gotcha)

Sonnet 5 (June 30; default for Free/Pro and in Claude Code) is Anthropic's most agentic Sonnet, and three architectural changes matter more than any benchmark. **(1) The 1M-token context is native and the only size** — no smaller variant, no beta flag, no long-context surcharge. **(2) Adaptive thinking is the default:** on Sonnet 4.6 a request without a thinking field ran without thinking; on Sonnet 5 the same request runs WITH adaptive thinking — the model decides how much reasoning each request deserves. **(3) Max output jumped to 128K tokens** — an entire refactored module or full report in one call.

The gotcha is the **new tokenizer**: Sonnet 5 tokenizes the same text into roughly **30% MORE tokens** than Sonnet 4.6. Per-token pricing looks familiar (\$2/\$10 per M introductory through Aug 31, then \$3/\$15), but an equivalent request can cost meaningfully more than the price sheet implies. The sharpest lesson of this wave: **price sheets are no longer comparable across models** — only cost-per-completed-task on your own golden set is honest. On capability, Sonnet 5 in some cases matches Opus 4.8, collapsing the mid/frontier boundary the 70/25/5 stack was built on.

The wave at a glance

Model	Status	Context	Pricing (per M)	Headline change
Claude Sonnet 5	Jun 30, GA	1M native / 128K out	\$2/\$10 promo, then \$3/\$15	New tokenizer (+~30% tokens); adaptive thinking default
Claude Fable 5	Jul 1 (redeploy)	Mythos-class	\$10/\$50 + usage credits	99%+ classifier gate; reroutes to Opus 4.8 in <5% of sessions
Claude Mythos 5	Restricted	Same model as Fable 5	Project Glasswing only	Safeguards lifted in specific domains; approved US orgs
GPT-5.6 Sol / Terra / Luna	Preview (~20 orgs)	Frontier family	\$5/\$30 · \$2.50/\$15 · \$1/\$6	Government-mediated rollout; Sol on Cerebras at 750 tok/s
Gemini 3.5 Pro	July window	2M tokens	TBA	Deep Think reasoning mode; missed June target

2. Fable 5 vs Mythos 5 — capability gating becomes the release architecture

Fable 5 and Mythos 5 are the **same underlying model**; the entire difference is the safeguard layer. That makes the pair the clearest example yet of **capability gating as a release architecture**: ship the intelligence to everyone, gate the dangerous capabilities by construction. Fable 5 is the public, safeguarded version (\$10/\$50 per M); Mythos 5 — safeguards lifted in specific domains — stays restricted to approved US organisations through Project Glasswing.

The mechanism is worth learning. A **classifier at better than 99% accuracy** sits in front of the model; when a query trips it (cyber-offence, bio, chem, model-distillation domains) the request is silently **rerouted to Opus 4.8** instead. Tuned conservatively, it fires in under 5% of sessions; the US government's AI standards body (CAISI) called the safeguards 'extraordinarily strong'. The architecture was battle-tested the hard way: after Amazon researchers found a jailbreak letting Fable 5 surface software vulnerabilities, Anthropic **pulled the model entirely on June 12 and redeployed July 1**. Economics changed too: 50% weekly limits through July 7, then **usage credits** — metered frontier access, not all-you-can-eat subscription.

The 19-day gap is the strongest availability argument for multi-provider routing you will ever see. Expect this pattern — classifier gate + fallback model + usage credits + revocable deployment — to become the standard frontier release template.

3. GPT-5.6 Sol, Terra, Luna — the tiered frontier and the government clock

OpenAI's answer is a three-tier family released as one wave. **Sol** is the flagship — long-horizon agentic work, software engineering, scientific research, cybersecurity — at \$5/\$30 per M. **Terra** delivers near-flagship capability at half the price (\$2.50/\$15). **Luna** is the fastest and cheapest (\$1/\$6), built for the high-volume classification and triage that dominates agent token budgets. Hardware note for agents: Sol launches on **Cerebras at up to 750 tokens/second** this July — an order of magnitude faster than typical frontier serving, which changes what 'interactive agent' means for multi-step loops.

The release process is the bigger story. GPT-5.6 is currently available to only **~20 government-vetted organisations via API and Codex** — not in ChatGPT at all — after the White House asked OpenAI to stage the rollout while national-security cybersecurity capability reviews complete. GA is promised in 'coming weeks'. Side by side, the pattern is unmistakable: Anthropic gates **capabilities** with classifiers, OpenAI gates **access** with a government-mediated preview, and the White House is drafting voluntary release standards with both labs plus Google. **Frontier releases are now a negotiated, government-adjacent process** — plan launches and dependencies around that clock.

4. The technique underneath — long-horizon agentic RL

Every model in this wave advertises the same phrase — 'long-horizon agentic' — and it comes from one training technique. Classic RLHF optimised single responses: one prompt, one output, one reward. Long-horizon agentic RL moves the optimisation target to the **cumulative return of entire trajectories** — multi-hour runs where the consequences of an early tool call propagate through hundreds of later steps. The model is no longer rewarded for a good answer; it is rewarded for a **completed task**, however many steps that takes. This is what produced Qwen3.7-Max's documented 35-hour, 1,158-tool-call autonomous run — and what Sol's 'long-horizon' claim rests on.

Mechanism	How it works	Why it matters
Hierarchical factorisation	Split the agent into a high-level planner + low-level executor; optimise with Hierarchical Advantage Estimation (e.g. HiPER)	Solves credit assignment: 'which of my 1,000 actions earned the reward?'
Process + outcome rewards	Auxiliary judges and programmatic rules score intermediate progress (right file found, test passes) on top of the final outcome	Dense signal instead of one sparse reward at the end of a 4-hour run
Behavioural reward bands	Progress rewards + efficiency bonuses (fewer searches, cheaper paths) + hard separation of successful vs failed trajectories	Why new models feel less 'thrashy' in tool loops

Practical implication for your own systems: **score trajectories, not just outputs** (Day 45's argument — now baked into how the models themselves are trained), and expect models tuned this way to keep going much longer before asking for help. Budget caps and loop detection matter **more**, not less.

5. Rebuilding the 70/25/5 routing stack for the new tiers

The routing framework survives; every tier assignment changes. Two disciplines before you re-route. **First, re-benchmark with the new tokenizer in the loop:** Sonnet 5's ~30% token inflation means a straight price-sheet comparison against 4.6 or Terra is wrong — run your 200-pair golden set and compare cost-per-successful-task (the Day 83 unit). **Second, treat availability as a routing input:** Fable 5 vanished for 19 days; GPT-5.6 is government-gated; classifier reroutes can silently swap the model answering your query. Multi-provider fallback chains are now the only way to promise your own users an SLO.

Tier	Workload	July 2026 defaults
Nano (~70% of calls)	Classification, extraction, triage	GPT-5.6 Luna \$1/\$6 · DeepSeek V4 Flash \$0.14/M · Gemini 3.5 Flash
Mid (~25%)	Code gen, structured reasoning, drafting, long-context synthesis	Claude Sonnet 5 \$2/\$10 promo (1M ctx absorbs the summariser role) · GPT-5.6 Terra \$2.50/\$15
Frontier (~5%)	Ambiguous planning, novel problems	Fable 5 \$10/\$50 (credits) · Sol \$5/\$30 (gated) · Opus 4.8 Fast \$10/\$50 · Gemini 3.5 Pro (July, 2M ctx + Deep Think)

Market signal: the frontier is now government-mediated — GPT-5.6 ships first to ~20 vetted orgs, Fable 5 returned only after CAISI-endorsed safeguards, and the White House is drafting voluntary release standards with OpenAI, Google, and Anthropic. Meanwhile tokenizer changes and usage-credit metering have made price sheets incomparable. Both point the same way: measure cost-per-successful-task on your own golden set, and architect for model substitution — the model answering your query can change (classifier reroute), be re-priced (credits), or disappear (19-day suspension) without your consent.

Viral app of the week — X Hosted MCP Server (api.x.com/mcp)

On June 30, X shipped a hosted Model Context Protocol server — one endpoint that lets Grok, Cursor, Claude Desktop, or any MCP client read X through your own developer credentials. **200+ tools:** search live posts, pull trends as they form, read full conversations, manage bookmarks. X's wedge over a GitHub or Notion MCP is the corpus — **the live public conversation**, something no other connector has. Read-only for now (the Write API isn't wired in, so no autonomous posting); the MCP layer is free but the API underneath is metered (~\$0.015 per post, \$0.20 with a link — priced to curb spam). The strategic read: the protocol Anthropic open-sourced in 2024 is now how a major consumer platform courts distribution — X is betting on **agent traffic over user growth**. When agents are the readers, being agent-callable is the new being indexed.

200+ tools exposed to any MCP client	Jun 30 hosted server went live	\$0.015 per post via the underlying X API
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Practical takeaways

- 1. Re-route on evidence, not price sheets.** Re-run your 200-pair golden set across the new tiers (Luna / Sonnet 5 / Terra / Fable 5) and compare cost-per-successful-task, not per-token price — Sonnet 5's tokenizer emits ~30% more tokens for the same text, so the \$2/\$10 promo (ends Aug 31) is cheaper than paper only if your tasks complete. Wire OTEL gen_ai spans before switching so the comparison is measured, not vibes.
- 2. Exploit the 1M-token default deliberately.** Native 1M context changes the RAG-vs-context math for repo-scale and long-trajectory work — but context engineering (Day 39) still wins: curated retrieval beats stuffing a million tokens of noise, and at 30% token inflation, stuffing is also expensive. Use the window for what only it can do: whole-codebase reasoning, full trajectory replays, long-document synthesis with adaptive thinking setting the effort.

3. Design for gated, revocable frontiers. Assume any frontier model can be suspended (Fable's 19 days), gated (GPT-5.6's preview), or silently rerouted (the <5% classifier fallback to Opus 4.8). Build multi-provider fallback chains with a tested degradation path to your mid tier, keep budget caps and loop detection tight, and watch the White House voluntary-release-standards announcement expected within days.

Tomorrow — Day 100: the milestone retrospective. '100 Days of Agentic AI' — grading the series' biggest calls (the compression thesis, the Aug 2 countdown, the IPO race, the protocol stack) and the 10 technologies that mattered most in 100 days, plus what to learn next.